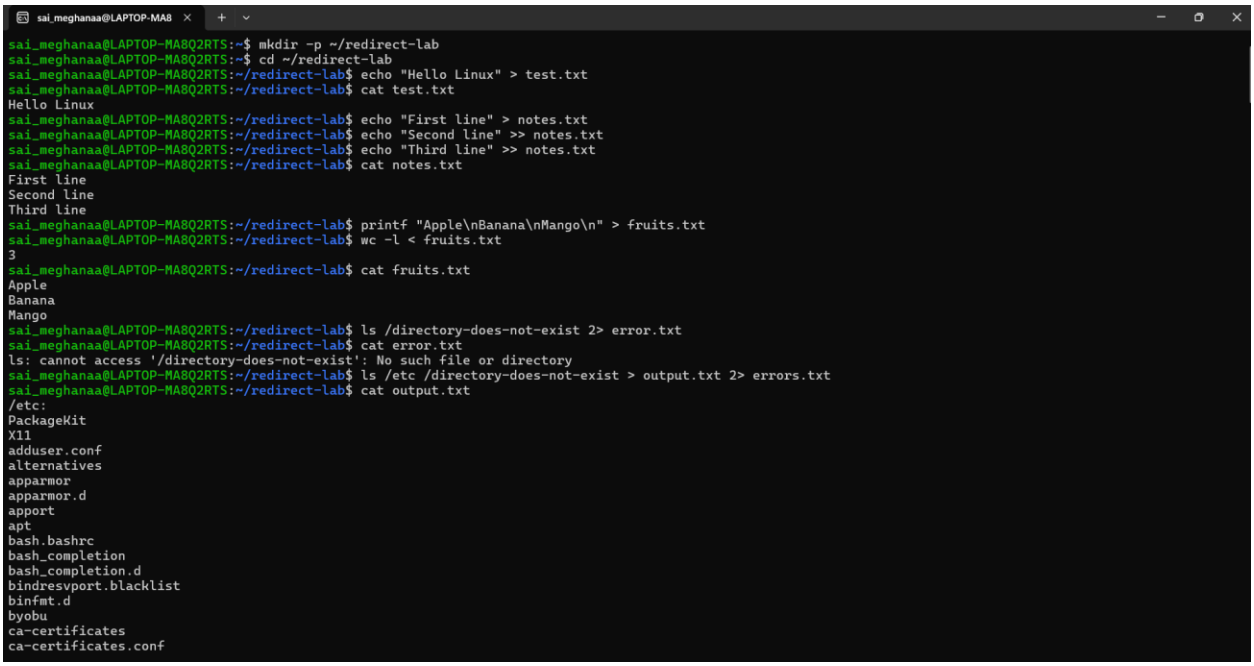
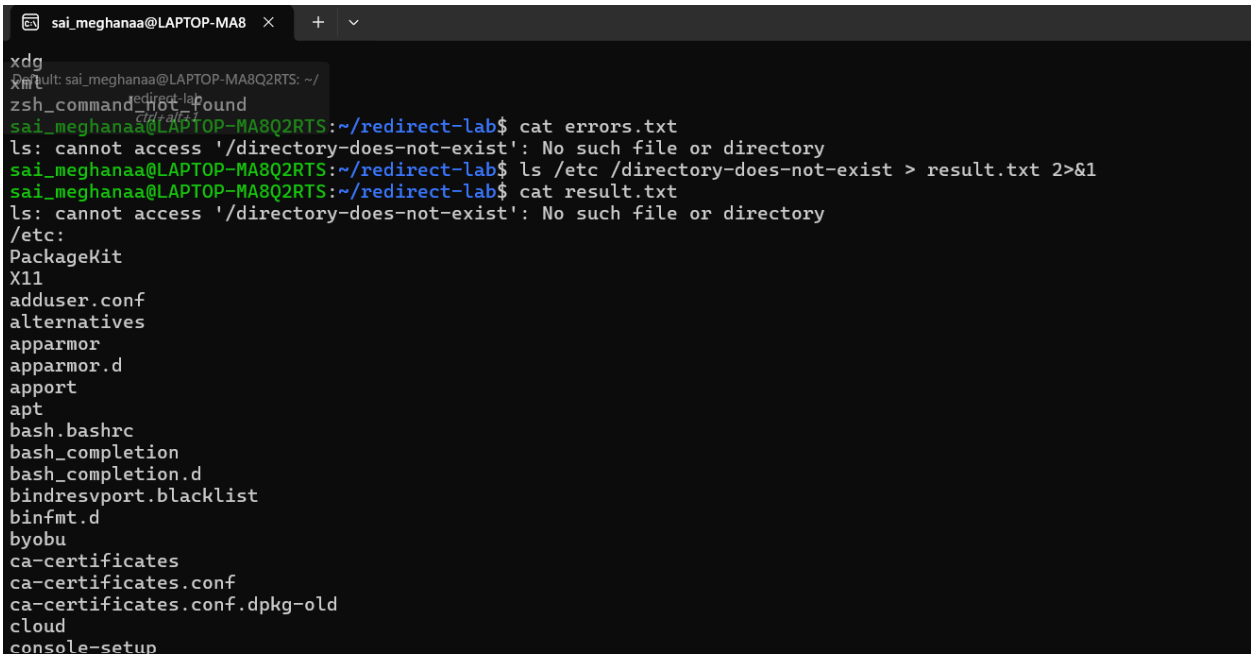
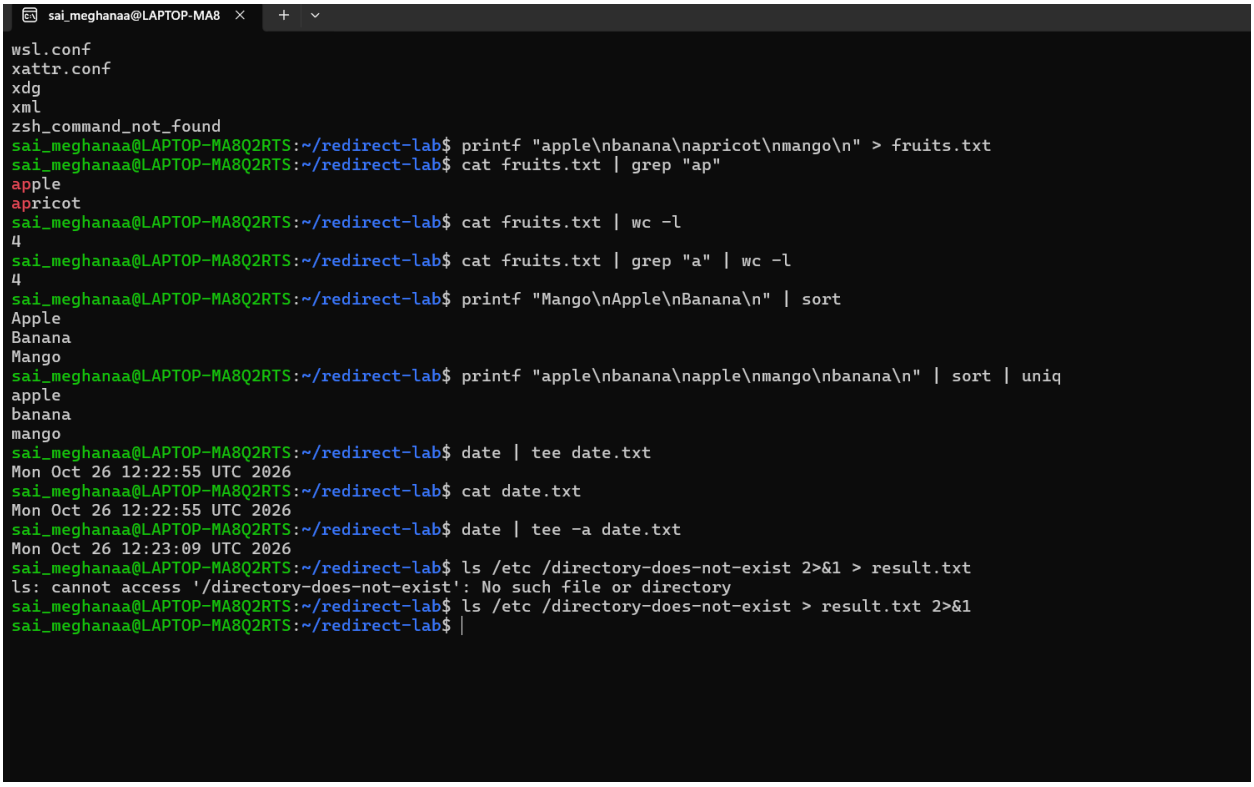
Linux Basic Commands:







Perform Data Analysis for 5 Products:

1. Scenario: An online shopping website is experiencing more product returns

- Descriptive Analysis — What happened? The number of products returned by customers has increased over the past month.
- Diagnostic Analysis — Why did it happen? Customers received damaged products, incorrect sizes, or items that did not match the product descriptions.
- Predictive Analysis — What might happen? Product returns and refund costs may continue to increase if product quality and delivery issues remain unresolved.
- Prescriptive Analysis — What should we do? Improve product quality checks, provide accurate product descriptions, and strengthen packaging and delivery processes.

2. Scenario: A college is experiencing a decline in student attendance

- Descriptive Analysis — What happened? Student attendance has decreased compared to the previous semester.
- Diagnostic Analysis — Why did it happen? Students may be facing transportation difficulties, losing interest in classes, or struggling to understand the subjects.
- Predictive Analysis — What might happen? Attendance may decline further, potentially affecting students' academic performance and examination results.
- Prescriptive Analysis — What should we do? Introduce interactive teaching methods, provide academic support, and identify students who need attendance counseling.

3. Scenario: A mobile phone company is receiving more customer complaints

- Descriptive Analysis — What happened? Customer complaints about mobile phones have increased in recent months.
- Diagnostic Analysis — Why did it happen? Customers are experiencing battery problems, software errors, and delayed customer support.
- Predictive Analysis — What might happen? Customer satisfaction and future sales may decrease if these problems continue.
- Prescriptive Analysis — What should we do? Improve software testing, address battery issues, and provide faster customer support and warranty services.

4. Scenario: A hospital is experiencing longer patient waiting times

- Descriptive Analysis — What happened? Patients are waiting longer than usual to consult doctors.
- Diagnostic Analysis — Why did it happen? The hospital has fewer available doctors during peak hours, inefficient appointment scheduling, and an increasing number of patients.
- Predictive Analysis — What might happen? Waiting times may increase further, leading to patient dissatisfaction and overcrowding.
- Prescriptive Analysis — What should we do? Optimize appointment scheduling, allocate more staff during peak hours, and introduce online appointment booking.

5. Scenario: A delivery company is experiencing delayed deliveries

- Descriptive Analysis — What happened? The percentage of packages delivered on time has decreased over the past few weeks.
- Diagnostic Analysis — Why did it happen? Traffic congestion, poor route planning, and an increase in package orders have caused delivery delays.
- Predictive Analysis — What might happen? Delivery delays and customer complaints may increase during busy periods if the problems continue.
- Prescriptive Analysis — What should we do? Use route optimization software, assign additional delivery personnel during peak periods, and track deliveries in real time.

Stock Market Data Analysis Using NumPy, Pandas and Matplotlib:

	A	B	C	D	E	F	G	H	I	J	K
1	Date	Company	Open	High	Low	Close	Volume				
2	1/1/2024	RELIANCE	1436.35	1626.49	1326.55	1506.11	23112				
3	1/2/2024	INFY	890.05	921.25	881.34	915.91	138393				
4	1/3/2024	TCS	2002.79	2144.4	1999.7	2140.05	83158				
5	1/4/2024	RELIANCE	2581.11	2623.58	2553.84	2566.63	364785				
6	1/5/2024	ICICIBANK	1260.61	1365.56	1195.82	1245.25	55580				
7	1/6/2024	ICICIBANK	2029.63	2057.53	1985.81	2012.09	231208				
8	1/7/2024	RELIANCE	1640.17	1797.21	1610.22	1706.38	25622				
9	1/8/2024	RELIANCE	1981.04	1990.33	1889.91	1907.03	124629				
10	1/9/2024	TCS	662.63	852.41	517.79	788.3	274951				
11	########	ICICIBANK	1261.53	1281.06	1158.9	1212.67	23912				
12	########	ICICIBANK	805.1	904.14	799.94	894.69	114248				
13	########	ICICIBANK	1146.95	1279.45	1100.19	1193.42	229949				
14	########	TCS	1866.78	1903.75	1721.34	1862.73	245514				
15	########	ICICIBANK	2848.75	3027.72	2759.07	3006.73	155852				
16	########	RELIANCE	721.23	760.43	714.45	729.41	407833				
17	########	TCS	1471.69	1525.96	1347.38	1411.09	376026				
18	########	HDFCBANK	1202.34	1310.88	1181.2	1285.23	188389				
19	########	INFY	686.38	883.76	570.54	632.78	91516				
20	########	TCS	513.81	676.9	407.78	603.97	410290				
21	########	INFY	2428.18	2442.99	2374.41	2382.36	63587				
22	########	RELIANCE	2657.76	2782.42	2608.13	2619.21	209191				
23	########	RELIANCE	1277.46	1342.5	1168.02	1279.26	198208				
24	########	INFY	2718.03	2812.47	2700.09	2780.24	326526				

Perform Mathematical Operations With and Without NumPy:

Untitled.ipynb | Untitled1.ipynb | sample_stock_market_data.cs | stock_data_with_risk.csv | water_consumption.csv

Delimiter: ,

	Date	Company	Open	High	Low	Close	Volume
1	1/1/2024	RELIANCE	1436.35	1626.49	1326.55	1506.11	23112
2	1/2/2024	INFY	890.05	921.25	881.34	915.91	138393
3	1/3/2024	TCS	2002.79	2144.4	1999.7	2140.05	83158
4	1/4/2024	RELIANCE	2581.11	2623.58	2553.84	2566.63	364785
5	1/5/2024	ICICIBANK	1260.61	1365.56	1195.82	1245.25	55580
6	1/6/2024	ICICIBANK	2029.63	2057.53	1985.81	2012.09	231208
7	1/7/2024	RELIANCE	1640.17	1797.21	1610.22	1706.38	25622
8	1/8/2024	RELIANCE	1981.04	1990.33	1889.91	1907.03	124629
9	1/9/2024	TCS	662.63	852.41	517.79	788.3	274951
10	1/10/2024	ICICIBANK	1261.53	1281.06	1158.9	1212.67	23912
11	1/11/2024	ICICIBANK	805.1	904.14	799.94	894.69	114248
12	1/12/2024	ICICIBANK	1146.95	1279.45	1100.19	1193.42	229949
13	1/13/2024	TCS	1866.78	1903.75	1721.34	1862.73	245514
14	1/14/2024	ICICIBANK	2848.75	3027.72	2759.07	3006.73	155852
15	1/15/2024	RELIANCE	721.23	760.43	714.45	729.41	407833
16	1/16/2024	TCS	1471.69	1525.96	1347.38	1411.09	376026
17	1/17/2024	HDFCBANK	1202.34	1310.88	1181.2	1285.23	188389
18	1/18/2024	INFY	686.38	883.76	570.54	632.78	91516
19	1/19/2024	TCS	513.81	676.9	407.78	603.97	410290

Untitled.ipynb | Untitled1.ipynb | sample_stock_market_data.cs | stock_data_with_risk.csv | water_consumption.csv

Code | Python 3 (ipykernel)

```
46 High Risk
```

[4]:
```
df.to_csv("stock_data_with_risk.csv", index=False)

print("CSV file saved successfully!")
```

```
CSV file saved successfully!
```

[5]:
```
import numpy as np

df["Risk"] = np.where(
    (df["High"] - df["Low"]) > 150,
    "High Risk",
    "Low Risk"
)

df.to_csv("stock_data_with_risk.csv", index=False)

print(df)
print("CSV file saved successfully!")
```

```
        Date    Company     Open     High      Low    Close   Volume  \
0   1/1/2024   RELIANCE  1436.35  1626.49  1326.55  1506.11    23112
1   1/2/2024       INFY   890.05   921.25   881.34   915.91   138393
2   1/3/2024        TCS  2002.79  2144.40  1999.70  2140.05    83158
3   1/4/2024   RELIANCE  2581.11  2623.58  2553.84  2566.63   364785
4   1/5/2024  ICICIBANK  1260.61  1365.56  1195.82  1245.25    55580
5   1/6/2024  ICICIBANK  2029.63  2057.53  1985.81  2012.09   231208
6   1/7/2024   RELIANCE  1640.17  1797.21  1610.22  1706.38    25622
7   1/8/2024   RELIANCE  1981.04  1990.33  1889.91  1907.03   124629
```

Untitled.ipynb | Untitled1.ipynb | sample_stock_market_data.cs | stock_data_with_risk.csv | water_cons

Code

[2]:
```
import pandas as pd

df = pd.read_csv("sample_stock_market_data.csv")
```

[3]:
```
risk = []

for i in range(len(df)):
    difference = df["High"].iloc[i] - df["Low"].iloc[i]

    if difference > 150:
        risk.append("High Risk")
    else:
        risk.append("Low Risk")


df["Risk"] = risk

print(df)
```

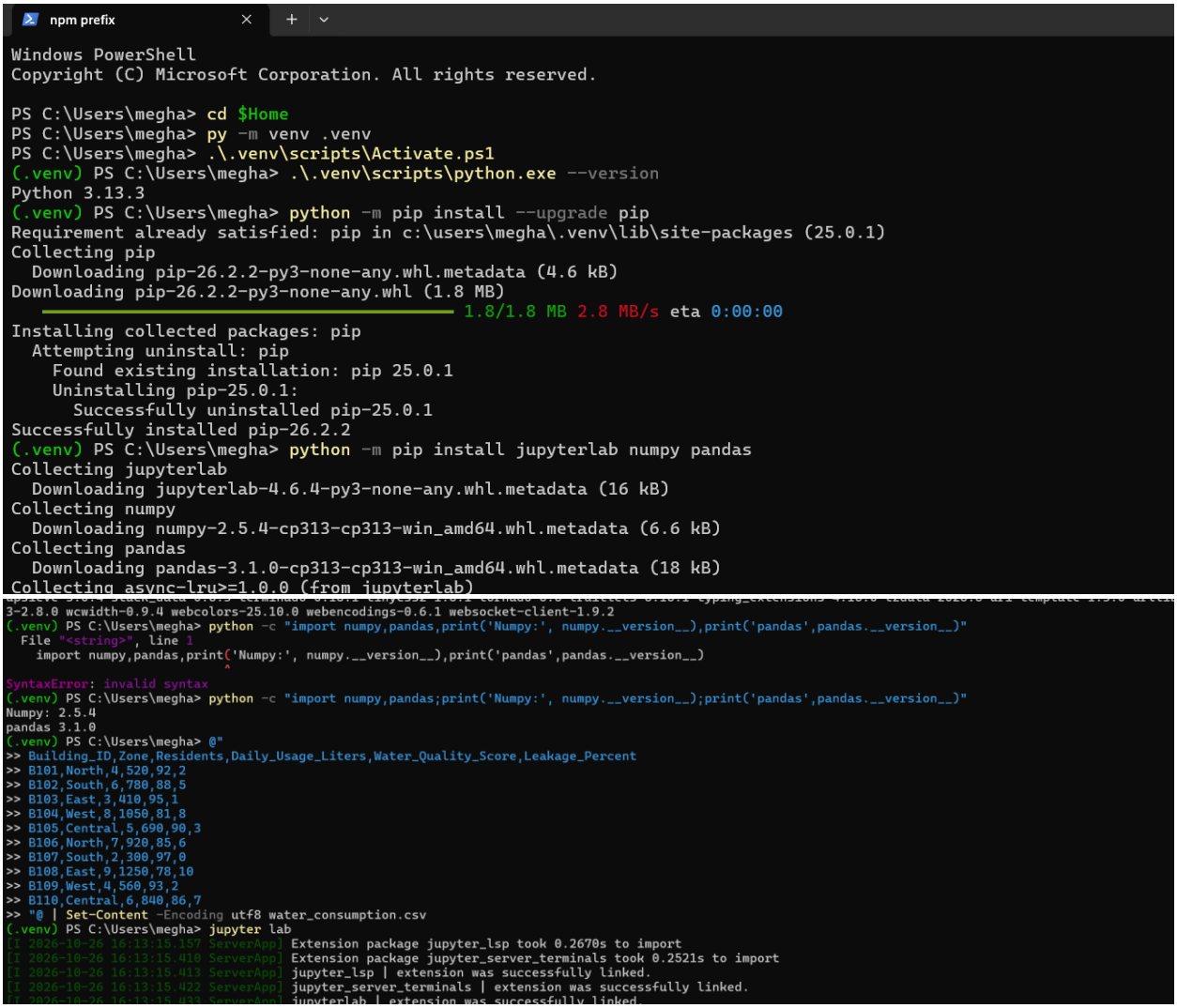
```
         Date    Company     Open     High      Low    Close  Volume  \
0    1/1/2024   RELIANCE  1436.35  1626.49  1326.55  1506.11   23112
1    1/2/2024       INFY   890.05   921.25   881.34   915.91  138393
2    1/3/2024        TCS  2002.79  2144.40  1999.70  2140.05   83158
3    1/4/2024   RELIANCE  2581.11  2623.58  2553.84  2566.63  364785
4    1/5/2024  ICICIBANK  1260.61  1365.56  1195.82  1245.25   55580
5    1/6/2024  ICICIBANK  2029.63  2057.53  1985.81  2012.09  231208
6    1/7/2024   RELIANCE  1640.17  1797.21  1610.22  1706.38   25622
7    1/8/2024   RELIANCE  1981.04  1990.33  1889.91  1907.03  124629
```

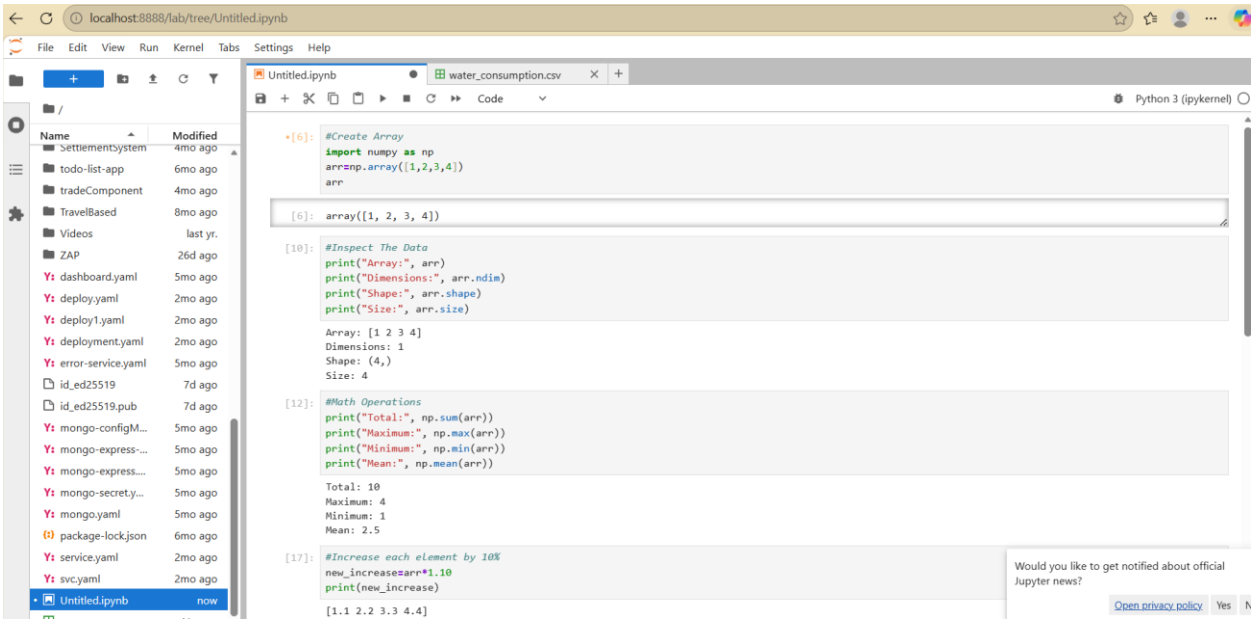
Untitled.ipynb | Untitled1.ipynb | sample_stock_market_data.cs | stock_data_with_risk.csv | water_consumption.csv

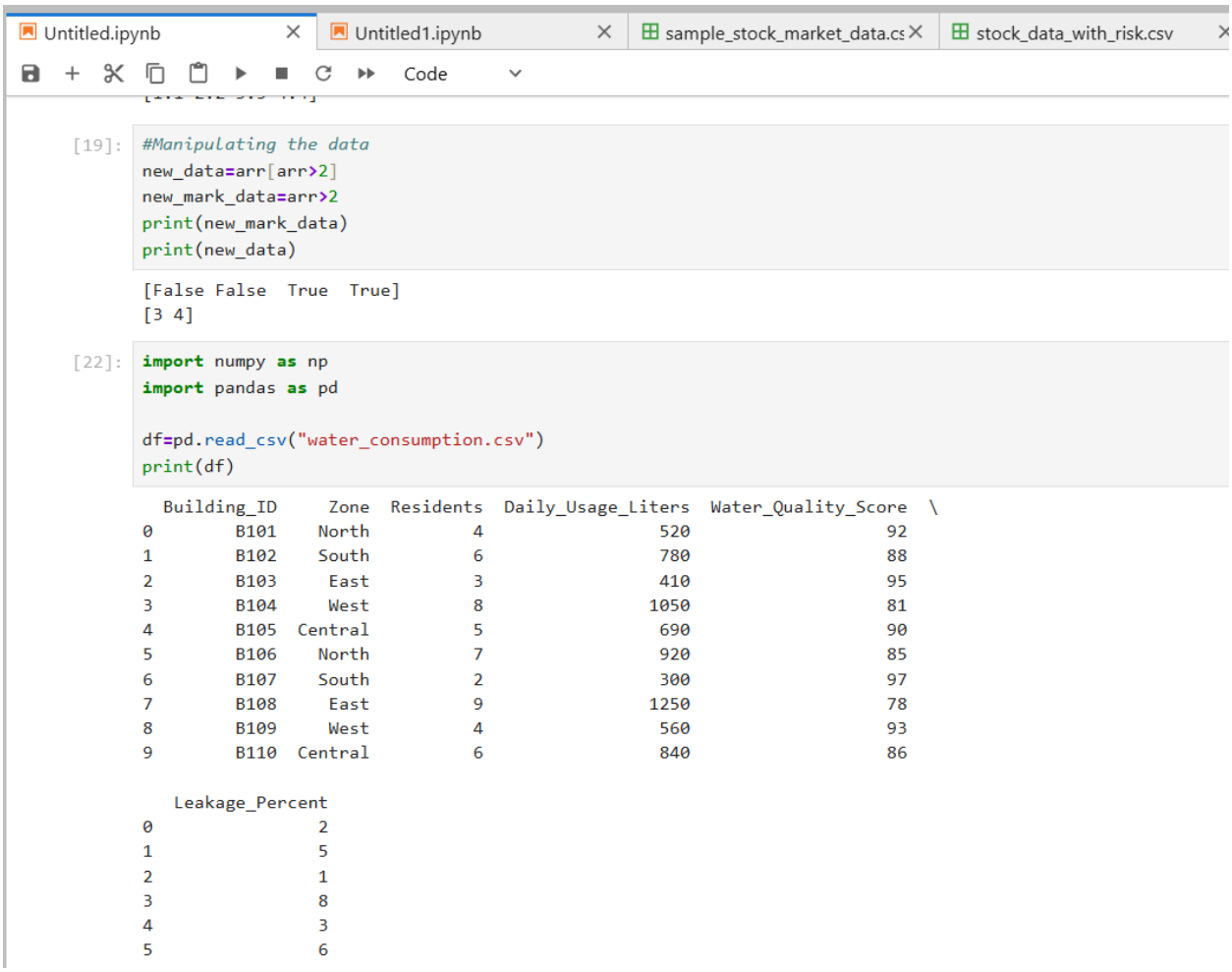
Delimiter: ,

	Date	Company	Open	High	Low	Close	Volume	Risk
1	1/1/2024	RELIANCE	1436.35	1626.49	1326.55	1506.11	23112	High Risk
2	1/2/2024	INFY	890.05	921.25	881.34	915.91	138393	Low Risk
3	1/3/2024	TCS	2002.79	2144.4	1999.7	2140.05	83158	Low Risk
4	1/4/2024	RELIANCE	2581.11	2623.58	2553.84	2566.63	364785	Low Risk
5	1/5/2024	ICICIBANK	1260.61	1365.56	1195.82	1245.25	55580	High Risk
6	1/6/2024	ICICIBANK	2029.63	2057.53	1985.81	2012.09	231208	Low Risk
7	1/7/2024	RELIANCE	1640.17	1797.21	1610.22	1706.38	25622	High Risk
8	1/8/2024	RELIANCE	1981.04	1990.33	1889.91	1907.03	124629	Low Risk
9	1/9/2024	TCS	662.63	852.41	517.79	788.3	274951	High Risk
10	1/10/2024	ICICIBANK	1261.53	1281.06	1158.9	1212.67	23912	Low Risk
11	1/11/2024	ICICIBANK	805.1	904.14	799.94	894.69	114248	Low Risk
12	1/12/2024	ICICIBANK	1146.95	1279.45	1100.19	1193.42	229949	High Risk
13	1/13/2024	TCS	1866.78	1903.75	1721.34	1862.73	245514	High Risk
14	1/14/2024	ICICIBANK	2848.75	3027.72	2759.07	3006.73	155852	High Risk
15	1/15/2024	RELIANCE	721.23	760.43	714.45	729.41	407833	Low Risk
16	1/16/2024	TCS	1471.69	1525.96	1347.38	1411.09	376026	High Risk
17	1/17/2024	HDFCBANK	1202.34	1310.88	1181.2	1285.23	188389	Low Risk
18	1/18/2024	INFY	686.38	883.76	570.54	632.78	91516	High Risk
19	1/19/2024	TCS	513.81	676.9	407.78	603.97	410290	High Risk
20	1/20/2024	INFY	2428.18	2442.99	2374.41	2382.36	63587	Low Risk

DataFrame Operations, Data Analysis and Data Manipulation Using NumPy,Pandas:




localhost:8888/lab/tree/Untitled.ipynb
File Edit View Run Kernel Tabs Settings Help
Untitled.ipynb
water_consumption.csv
Code
Python 3 (ipykernel)
Name
Modified
todo-list-app 6mo ago
tradeComponent 4mo ago
TravelBased 8mo ago
Videos last yr.
ZAP 26d ago
dashboard.yaml 5mo ago
deploy.yaml 2mo ago
deploy1.yaml 2mo ago
deployment.yaml 2mo ago
error-service.yaml 5mo ago
id_ed25519 7d ago
id_ed25519.pub 7d ago
mongo-configM... 5mo ago
mongo-express-... 5mo ago
mongo-express.... 5mo ago
mongo-secret.y... 5mo ago
mongo.yaml 5mo ago
package-lock.json 6mo ago
service.yaml 2mo ago
svc.yaml 2mo ago
Untitled.ipynb now
[6]: #Create Array
import numpy as np
arr=np.array([1,2,3,4])
arr
[6]: array([1, 2, 3, 4])
[10]: #Inspect The Data
print("Array:", arr)
print("Dimensions:", arr.ndim)
print("Shape:", arr.shape)
print("Size:", arr.size)
Array: [1 2 3 4]
Dimensions: 1
Shape: (4,)
Size: 4
[12]: #Math Operations
print("Total:", np.sum(arr))
print("Maximum:", np.max(arr))
print("Minimum:", np.min(arr))
print("Mean:", np.mean(arr))
Total: 10
Maximum: 4
Minimum: 1
Mean: 2.5
[17]: #Increase each element by 10%
new_increase=arr*1.10
print(new_increase)
[1.1 2.2 3.3 4.4]
Would you like to get notified about official Jupyter news?
Open privacy policy
Yes


Untitled.ipynb
Untitled1.ipynb
sample_stock_market_data.cs
stock_data_with_risk.csv
Code
[19]: #Manipulating the data
new_data=arr[arr>2]
new_mark_data=arr>2
print(new_mark_data)
print(new_data)
[False False True True]
[3 4]
[22]: import numpy as np
import pandas as pd
df=pd.read_csv("water_consumption.csv")
print(df)
Building_ID Zone Residents Daily_Usage_Liters Water_Quality_Score \
0 B101 North 4 520 92
1 B102 South 6 780 88
2 B103 East 3 410 95
3 B104 West 8 1050 81
4 B105 Central 5 690 90
5 B106 North 7 920 85
6 B107 South 2 300 97
7 B108 East 9 1250 78
8 B109 West 4 560 93
9 B110 Central 6 840 86
Leakage_Percent
0 2
1 5
2 1
3 8
4 3
5 6

Untitled.ipynb | Untitled1.ipynb | sample_stock_market_data.cs | stock_data_with_risk.csv

Code

[27]:
```
#Inspect The Data
print("Dimensions:", df.ndim)
print("Shape:", df.shape)
print("Size:", df.size)

print('Info:',df.info())
print('Describe:',df.describe())
print('Is Null:',df.isnull())
print('Is Null Count:',df.isnull().sum())
```

```
Dimensions: 2
Shape: (10, 6)
Size: 60
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 6 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Building_ID          10 non-null     str
 1   Zone                 10 non-null     str
 2   Residents            10 non-null     int64
 3   Daily_Usage_Liters   10 non-null     int64
 4   Water_Quality_Score  10 non-null     int64
 5   Leakage_Percent      10 non-null     int64
dtypes: int64(4), str(2)
memory usage: 612.0 bytes
Info: None
Describe:        Residents  Daily_Usage_Liters  Water_Quality_Score  Leakage_Percent
count  10.000000           10.000000            10.000000        10.000000
mean    5.400000          732.000000            88.500000         4.400000
std     2.221111          295.401798             6.096447         3.306559
min     2.000000          300.000000            78.000000         0.000000
25%     4.000000          530.000000            85.250000         2.000000
```

Untitled.ipynb | Untitled1.ipynb | sample_stock_market_data.cs | stock_data_with_risk.csv | water_consumption.csv

Code | Python 3 (ipykern

```
Average: 732.0
```

[33]:
```
df["Daily_Usage_Liters"].to_numpy
```

[33]:
```
<bound method IndexOpsMixin.to_numpy of 0     520
1     780
2     410
3    1050
4     690
5     920
6     300
7    1250
8     560
9     840
Name: Daily_Usage_Liters, dtype: int64>
```

[37]:
```
print(np.sum(df["Daily_Usage_Liters"].to_numpy(),axis=0))
```

```
7320
```

[38]:
```
df.head()
```

[38]:

	Building_ID	Zone	Residents	Daily_Usage_Liters	Water_Quality_Score	Leakage_Percent
0	B101	North	4	520	92	2
1	B102	South	6	780	88	5
2	B103	East	3	410	95	1
3	B104	West	8	1050	81	8
4	B105	Central	5	690	90	3

[39]:
```
df.tail(4)
```

Untitled.ipynb | Untitled1.ipynb | sample_stock_market_data.cs | stock_data_with_risk.csv | water_consumption.csv

Code | Python 3 (ipyker

[39]:
```
df.tail(4)
```

[39]:

	Building_ID	Zone	Residents	Daily_Usage_Liters	Water_Quality_Score	Leakage_Percent
6	B107	South	2	300	97	0
7	B108	East	9	1250	78	10
8	B109	West	4	560	93	2
9	B110	Central	6	840	86	7

[41]:
```
print(df.columns.tolist())
```

```
['Building_ID', 'Zone', 'Residents', 'Daily_Usage_Liters', 'Water_Quality_Score', 'Leakage_Percent']
```

[42]:
```
print(df.info())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 6 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Building_ID          10 non-null     str
 1   Zone                 10 non-null     str
 2   Residents            10 non-null     int64
 3   Daily_Usage_Liters   10 non-null     int64
 4   Water_Quality_Score  10 non-null     int64
 5   Leakage_Percent      10 non-null     int64
dtypes: int64(4), str(2)
memory usage: 612.0 bytes
None
```

[43]:
```
df["Increased_Usage"] = df["Daily_Usage_Liters"] * 1.10
```

Untitled.ipynb | Untitled1.ipynb | sample_stock_market_data.cs | stock_data_with_risk.csv | water_consumpt

Code

None

[43]:
```
df["Increased_Usage"] = df["Daily_Usage_Liters"] * 1.10

df[["Building_ID", "Daily_Usage_Liters", "Increased_Usage"]]
```

[43]:

	Building_ID	Daily_Usage_Liters	Increased_Usage
0	B101	520	572.0
1	B102	780	858.0
2	B103	410	451.0
3	B104	1050	1155.0
4	B105	690	759.0
5	B106	920	1012.0
6	B107	300	330.0
7	B108	1250	1375.0
8	B109	560	616.0
9	B110	840	924.0

[44]:
```
high_usage = df[df["Daily_Usage_Liters"] > 800]
high_usage
```

[44]:

	Building_ID	Zone	Residents	Daily_Usage_Liters	Water_Quality_Score	Leakage_Percent	Increased_Usage
3	B104	West	8	1050	81	8	1155.0
5	B106	North	7	920	85	6	1012.0
7	B108	East	9	1250	78	10	1375.0

Mode: Command | Cell 1

Untitled.ipynb | Untitled1.ipynb | sample_stock_market_data.cs | stock_data_with_risk.csv

Code

[45]:
```
high_leakage = df[df["Leakage_Percent"] > 5]
high_leakage
```

[45]:

	Building_ID	Zone	Residents	Daily_Usage_Liters	Water_Quality_Score	Leakage_Percent	Increased_Usage
3	B104	West	8	1050	81	8	1155.0
5	B106	North	7	920	85	6	1012.0
7	B108	East	9	1250	78	10	1375.0
9	B110	Central	6	840	86	7	924.0

[46]:
```
df[(df["Residents"] > 5) & (df["Daily_Usage_Liters"] < 900)]
```

[46]:

	Building_ID	Zone	Residents	Daily_Usage_Liters	Water_Quality_Score	Leakage_Percent	Increased_Usage
1	B102	South	6	780	88	5	858.0
9	B110	Central	6	840	86	7	924.0

[47]:
```
df["Estimated_Wastage_Liters"] = (
    df["Daily_Usage_Liters"] * df["Leakage_Percent"] / 100
)

df[["Building_ID", "Daily_Usage_Liters",
    "Leakage_Percent", "Estimated_Wastage_Liters"]]
```

[47]:

	Building_ID	Daily_Usage_Liters	Leakage_Percent	Estimated_Wastage_Liters
0	B101	520	2	10.4
1	B102	780	5	39.0

Untitled.ipynb | Untitled1.ipynb | sample_stock_market_data.cs | stock_data_with_risk.csv

Code

2	B103	410	1	4.1
3	B104	1050	8	84.0
4	B105	690	3	20.7
5	B106	920	6	55.2
6	B107	300	0	0.0
7	B108	1250	10	125.0
8	B109	560	2	11.2
9	B110	840	7	58.8

[48]:
```
wastage = df["Estimated_Wastage_Liters"]

print("Total estimated wastage:", wastage.sum())
print("Average wastage:", wastage.mean())
print("Minimum wastage:", wastage.min())
print("Maximum wastage:", wastage.max())
```

```
Total estimated wastage: 408.4
Average wastage: 40.839999999999996
Minimum wastage: 0.0
Maximum wastage: 125.0
```

[ ]: